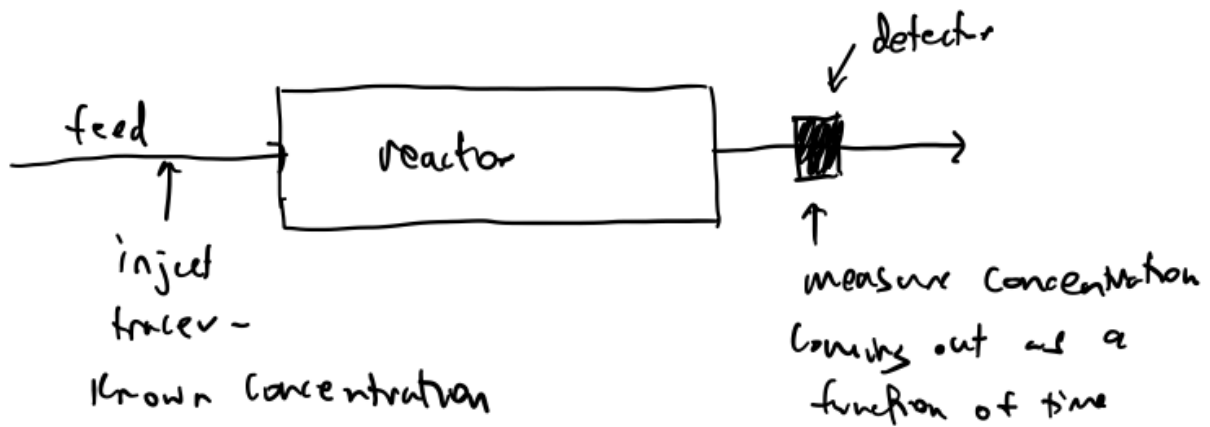


30 Pulse and step experiments

Residence time distributions (RTDs) are measured by dosing a "tracer" to the reactor.



A tracer

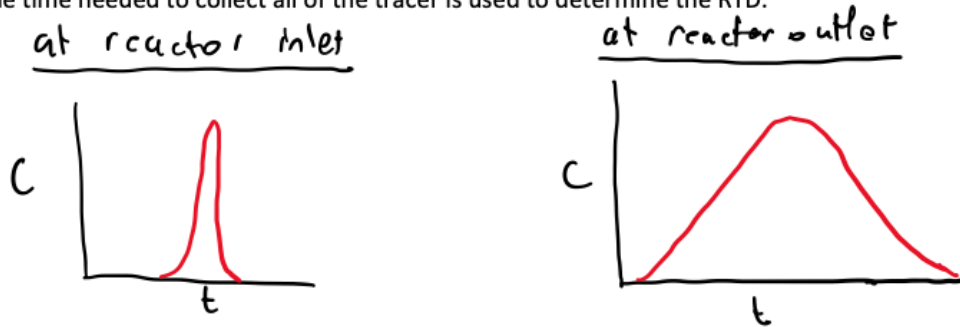
- Is non-reactive
- Is easily detectable

- Has physical properties similar to the reactant mixture & is completely soluble in the reactant/product mixture
- Is often a colored material, a radioactive material, or an inert gas

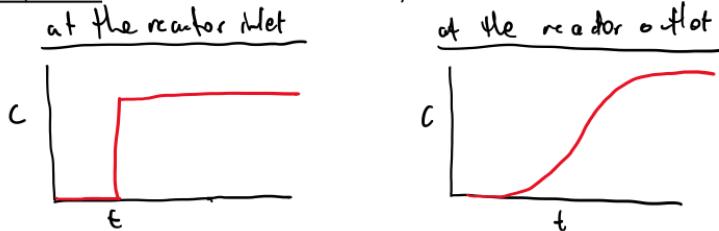
The tracer can be dosed in either a pulse or step experiment.

Pulse experiment: An amount of tracer is suddenly injected in one shot

The time needed to collect all of the tracer is used to determine the RTD.



Step experiment: A concentration of tracer is constantly fed to the reactor



$$C_{in}(t) = \begin{cases} 0 & t < 0 \\ C_0 & t \geq 0 \end{cases}$$

The time needed for the concentration of tracer exiting the reactor to equal that entering the reactor is used to determine the RTD.

Important RTD statistics:

- ΔN : Amount of tracer leaving the reactor between t and Δt :

$$\Delta N = C(t) v \Delta t$$

from the $C(t)$ curve @ the reactor outlet

- $\frac{\Delta N}{N_0}$: The fraction of tracer/reactor contents that has a residence time between t and Δt :

$$\frac{\Delta N}{N_0} = \frac{v C(t) \Delta t}{N_0} \quad / \quad \begin{array}{l} N_0 \text{ is the total amount} \\ \text{that was injected} \end{array}$$

$$N_0 = \int_0^{\infty} v C(t) dt$$

- $E(t)$: Residence time distribution

$$E(t) = \frac{\frac{\Delta N}{N_0}}{\Delta t} = \frac{v C(t) \Delta t}{N_0 \Delta t} = \frac{v C(t)}{N_0} = \frac{v C(t)}{\int_0^{\infty} v C(t) dt}$$

if v is constant, $E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt}$

The fraction of tracer/reactor contents leaving the reactor that has resided in the reactor for a time between t_1 and $t_2 = \int_{t_1}^{t_2} E(t) dt$

Also, $\int_0^{\infty} E(t) dt = 1$

- $F(t)$: Cumulative distribution function. Used for step experiments.

$$F(t) = \int_0^t E(t) dt, \quad F(t) \text{ is the fraction of tracer / reactor contents exiting that has been in the reactor for less than time } t$$

- t_m : Mean residence time

$$t_m = \frac{\int_0^{\infty} t E(t) dt}{\int_0^{\infty} E(t) dt} = \int_0^{\infty} t E(t) dt$$

$\nwarrow$ = to 1 by definition

- σ^2 : Variance. Provides an indication of the "spread" in the data

$$\sigma^2 = \int_0^{\infty} (t - t_m)^2 E(t) dt$$